

Question number	Answer	Mark
4(a)	The only correct answer is C <i>A is incorrect because cellulose is made of β glucose</i> <i>B is incorrect because cellulose is made of β glucose</i> <i>D is incorrect because cellulose is a straight chain polymer</i>	(1)

Question number	Answer	Mark																								
4(b)	<table><tr><th rowspan="2">Structure</th><th colspan="4">Microorganism</th></tr><tr><th>Bacteria only</th><th>Fungi only</th><th>both bacteria and fungi</th><th>neither bacteria nor fungi</th></tr><tr><td>mitochondria</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr><tr><td>nuclei</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr><tr><td>70S (small) ribosomes</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr></table>	Structure	Microorganism				Bacteria only	Fungi only	both bacteria and fungi	neither bacteria nor fungi	mitochondria	[x]	[x]	[x]	[x]	nuclei	[x]	[x]	[x]	[x]	70S (small) ribosomes	[x]	[x]	[x]	[x]	(3)
Structure	Microorganism																									
	Bacteria only	Fungi only	both bacteria and fungi	neither bacteria nor fungi																						
mitochondria	[x]	[x]	[x]	[x]																						
nuclei	[x]	[x]	[x]	[x]																						
70S (small) ribosomes	[x]	[x]	[x]	[x]																						

Question number	Answer	Mark
4(c)(i)	The only correct answer is D <i>A is incorrect because relative abundance of Streptococcus is highest in cubs, Escherichia and Shigella highest in sub-adults and Clostridium highest in adults</i> <i>B is incorrect because relative abundance of Streptococcus is highest in cubs, Escherichia and Shigella highest in sub-adults and Clostridium highest in adults</i> <i>C is incorrect because relative abundance of Streptococcus is highest in cubs, Escherichia and Shigella highest in sub-adults and Clostridium highest in adults</i>	(1)

Question number	Answer	Additional guidance	Mark
4(c)(ii)	A description that includes four of the following points: • source of sample identified (1) • name of method of determining numbers (1) • credit appropriate detail of method (1) • method for {identifying / eliminating} bacteria (1) • description of how the relative abundance will be determined (1)	e.g. faeces, intestinal flush IGNORE other samples e.g. colony counts, dilution plating, colorimeter, spectrophotometer, turbidity, counting chamber, haemocytometer ACCEPT description of the method e.g. light passed through to measure transmission e.g. serial dilutions, shaking sample before counting, safety precautions, culture conditions, aseptic technique e.g. colour of colonies, use of antibiotics, use of antibodies, use of {selective / indicator} media e.g. number of each streptococcus in each sample from each age group of panda comparing light transmission for each age group percentage of streptococcus of total number of bacteria	(4)

